

Operating Systems

INTRODUCTION

Course Instructors

Ms. Sidra Khalid – Course Coordinator

- **Office- C-505G**
- **Email: sidra.khalid@nu.edu.pk**
- **Office Hours: Monday, Friday: 12.00pm-2.00pm, Tuesday 1.00pm-2.30pm**

Dr. Ali Zeeshan Ijaz

- **Office : C-503G**
- **Email : ali.zeeshan@isb.nu.edu.pk**
- **Office hours : Wed - 11-12PM, Thurs - 2-3PM, Fri 10-11AM**

Course Instructors

Ms. Sana Razzaq

- **Office- C-504B**
- **Email: sana.razzaq@nu.edu.pk**
- **Office hours : Tuesday - 11:30-1:00PM, Wednesday - 10:00-11:30 Thurs - 11:30-1:00PM**

Ms. Rabail Zahid

- **Office- C-204A,**
- **Email: rabail.zahid@isb.nu.edu.pk**
- **Office hours: Tuesday, Thursday: 10-11Friday: 10-11**

Display

LMS - Google Classroom

taf3k42

Your Introduction!

Your name

Your Interests or Skills or Achievements

Your motivation for taking this class/what you want to learn

Class Rules

No Compromise on Classroom Discipline

- Bring your own pen and a register/copy/paper.
- No Food, Tea, coffee.
- No Unnecessary IN/OUT from Lecture
- Mobile phones shall be OFF/SILENT
  - RS 2000/- Fine.
- No gossips/talks during Lecture.

Some Rules

- Never ever miss a class
- Never ever “sleep” in the class
- **Never use mobile phones in the class**

- Above all, whatever you do, please do not disturb others

Course Policy

- Assignments will be submitted online on Google Classroom.
- All quizzes will be both announced and unannounced.
- No Retake of assignments, project or quizzes.
- **Missed or late assignment and Project will be marked zero.**
- **Missed Demo of assignment and Project will be marked zero.**
- **No Codebase evaluation will be done. Demo is mandatory.**

Dishonesty, Plagiarism

You can fool some of the people all of the time, and all of the people some of the time,

but you can not fool all of the people all of the time.

Abraham Lincoln,
16th president of US (1809 - 1865)

Dishonesty, Plagiarism

- All parties involved in any kind of cheating in any exam (Quizzes, Assignments) will get 0 marks.
- Cheating in Project case will be forwarded to DC for Action.
- Cheating in exams(sessional I and II, and Final exams) will earn F Grade in course.

Marks Distribution

Assessment Type	Weight
Quizzes	10
Assignments –(2)	8
Project	12
Sessional Exams 1	15
Sessional Exams 2	15
Final Exam	40

Formula for GOOD GRADES

All of you have **A+**

Just do the following

- Maintain Attendance
- Positive class attitude and Participation
- Perform all class task
- Do Quizzes , Assignments, and Project by YOURSELF
- Follow all Rules!

And keep your ***A+***

Pre-Requisites

- Must have passed Data Structures
- Familiar with programming C/C++

Topics

- Introduction
- Dockers
- Processes
- inter-process communication
- threads
- CPU Scheduling
- Synchronization
- Semaphores
- Deadlocks
- Memory Management
- Advance Topics in OS
- …

Books

- Text Book:
  - Operating System Concepts, 9th Editon, Silberschatz Galvin Gagne
- Reference Books:
  - Modern Operating Systems, 3rd Edition, Andrew S. Tanenbaum

What is an Operating System?

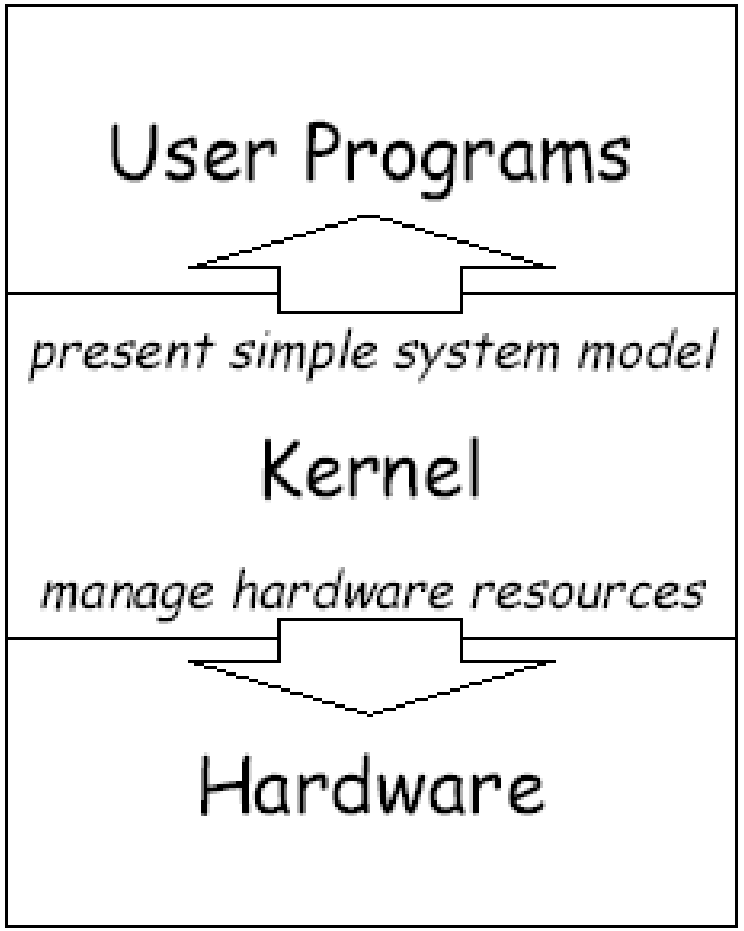
- It is a program!
- Top-down view
  - Provides an extended or virtual machine abstraction to user programs
    - Easier to program than the underlying hardware.
    - All services are invoked and accomplished through system calls.
- Bottom-up view
  - Acts as a resource manager of a complex system
  - Resources consist of processors, memories, timers, disks, keyboard, network interfaces, printers etc.
  - OS manages allocation of these resources to user programs in an orderly and controlled manner

What is an Operating System?

- An operating system is a program that manages a computer's hardware
- Act as an intermediary between users and the hardware
- OS do nothing by themselves.
- Similar to subroutine libraries, do nothing unless they are invoked by programs

Two Goals of Operating Systems

- Manage hardware resources
  - System operates smoothly: efficiently, reliably, securely
- Present abstract system model to programmer
  - Simple and convenient access to and control of resources



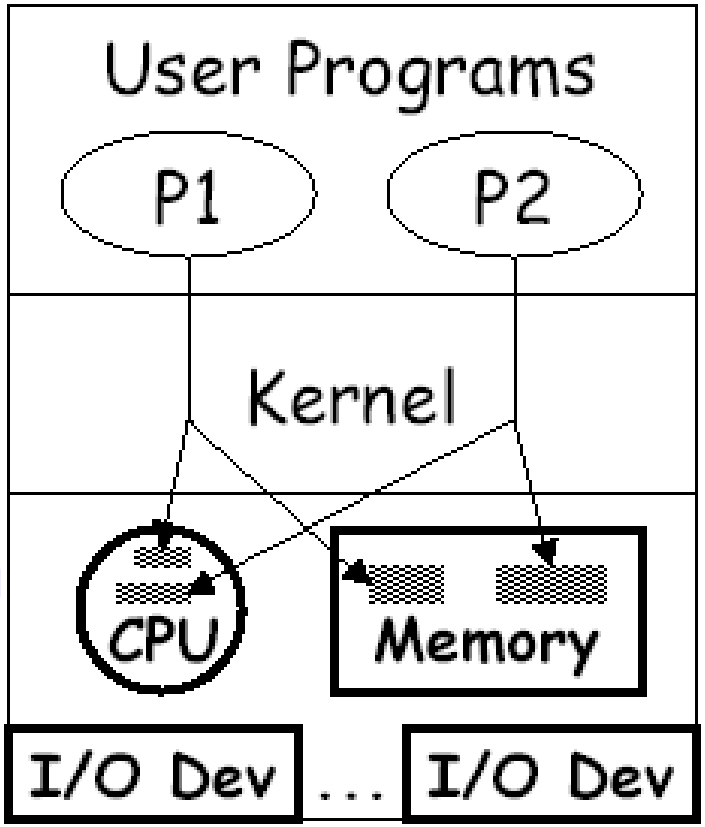
Manager/Coordinator of Resources

Coordinates who gets what

- "who": running programs
- "what": resources
- "when": scheduling time
- "where": organizing space
- "whether": limits, rights

Goal: smooth system operation

- efficiency,reliability,security

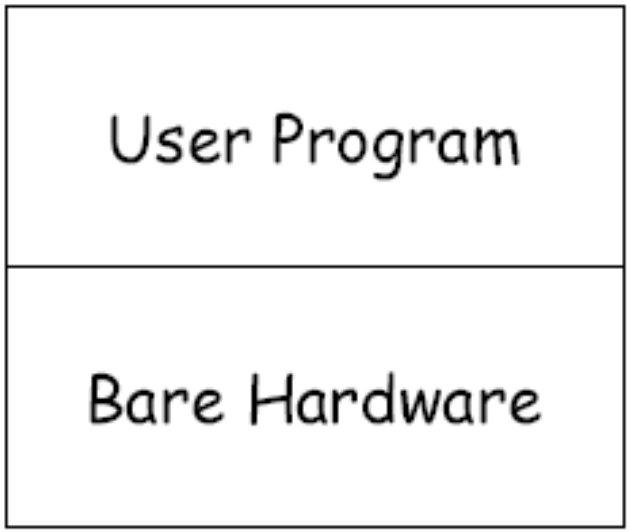


Resource multiplexing

- OS multiplexes resources in two ways:
- In time
  - Time multiplexing involves different programs taking turns in using the resource.
    - Example: CPU scheduling
    - printer sharing.
- In space
  - Space multiplexing involves different program getting part of the resource possibly at the same time.
    - Example: memory is divided into several running programs.

What If No Operating System?

- All we have is bare hardware
- You want to run a program
  - How do you load it?
  - How do you run it?
  - What happens when it completes?
- Need at least some minimal OS to do these functions



…Lets go back to the “stone age” of computing…

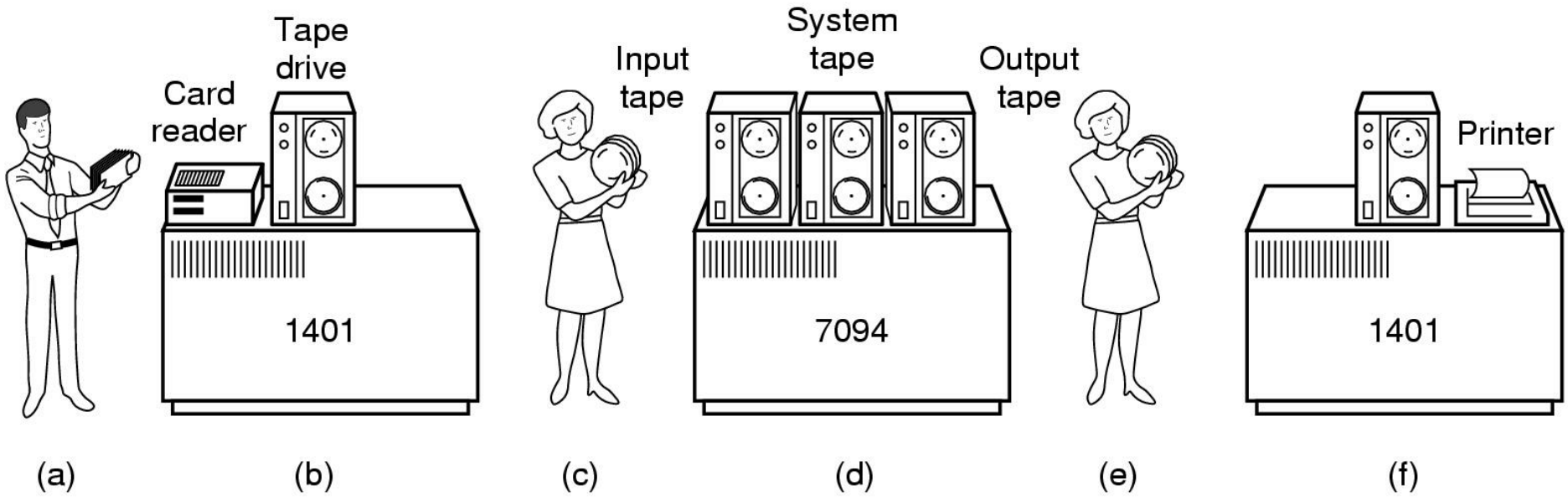
Early Systems

- No Operating systems,
- your Data and your Programs
  - X = X + 1;
- As
  - Load Register1, x
  - Load Direct Register2, 1
  - Add Register3, Register2, Register1
  - Store Register3, x
- Time slots allocated to scientists
- Enter programs using binary switches

Early Systems

- Problem:
  - computer idles while programmer sets things up.
  - Poor utilization of huge investment.
- Solution:
  - Hire a specialized person to do setup.
- Problem:
  - Faster than programmer, but still a lot slower than the machine.
- Solution:
  - Build a batch monitor.

The Monitor



Early batch system

- bring cards to 1401
- read cards to tape
- put tape on 7094 which does computing
- put tape on 1401 which prints output

Simple Batch System

- The user submits a job (written on a card or tape) to a computer operator.
- The computer operator place a batch of several jobs on a input device
- A special program, the monitor, manages the execution of each program in the batch
- Resident monitor is in the main memory and available for execution

The Monitor

- Monitor reads job one at a time from the input device
- Monitor places a job in the user program area
- A monitor instruction branches to the start of the user program
- Execution of the user program continues until:
  - End of program occurs
  - Error occurs

Why study operating systems?

- Point of describing change isn't "Look how stupid batch processing is" – it was right for tradeoffs of the time, but not anymore
- Point is: have to change with changing technology
- Situation today is much like it was in the late 60's OS's today are enormous, complex things
  - 100k's of lines (or >1M lines)
  - Windows NT is 20M lines
  - Windows2000 and Windows XP are about 40M lines.
- Key aspect of this course, understand OS's so we can simplify them!

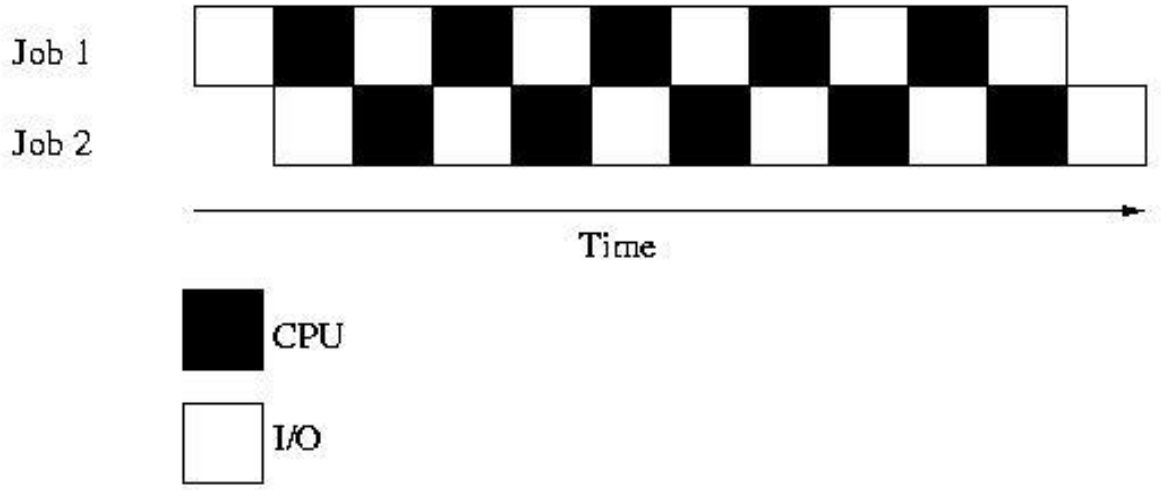
Beyond batch processing systems

- The Era of Timesharing and Multiprogramming

- Multiprogramming - Execute multiple jobs simultaneously
- Timesharing - Processor's time is shared among multiple users

Multiprogramming

- Execute multiple jobs simultaneously
- But a CPU can execute a single instruction at a time???
- One job can use the CPU while the other is waiting for I/O



- Small jobs not delayed by large jobs
- Context switching

Time Sharing System (TSS)

- Processor's time is shared among multiple users
- Use cheap terminals to let multiple users interact with the system at the same time.
- OS does timesharing to give illusion of each user has own computer

MultiProcessing vs MultiCore

- As processors became less expensive, became common to have multiple processors in a single computer.
- Multiprocessor systems contain multiple processors in separate packages
- Multicore systems have multiple processors in a single package
- Multiprocessor/multicore systems require specific support from the operating system
  - Coordinating access to shared data-structures within the operating system becomes much trickier
  - Process scheduling also takes multiprocessor systems into account to maximize cache utilization

- **Kinds of Operating system**

Operating systems for Personal Computing

- Computers are cheap, so give everyone a computer.
- Initially, OS became a subroutine library again (MSDos)
- Maximizing hardware utilization is less important – responding to user interaction is top priority!
- Much more code is devoted to making the computer easy to use
- Important to provide a simplified, user-friendly user interface
- Since then, adding back in memory protection, multiprogramming, etc.

Operating systems for Distributed Computing

- Computers so cheap – give people a bunch of them
  - I have a 2 PCs at home, 1 in my office, 3 smart phones and share some machines in a lab how do I coordinate a bunch of machines?
- Networks fast – allow machines to share resources and data easily
- Networks cheap – allow geographically distributed machines to interact

Mainframe and server operating systems

- maximize utilization of hardware
- Operating system doesn't require a graphical user interface
- Rather, must support very efficient handling of I/O, and possibly scheduling of many processes

Operating systems for Mobile/tablet devices

- Must be responsive and user-friendly, like PC operating systems
  - But, must also try to maximize battery life through careful hardware resource management.
- With smartphones, must support download, installation, execution, and uninstallation of wide range of applications •
  - But, basic device capabilities (e.g. voice calls, SMS) must also be rock-solid reliable
- Must support intermittent connectivity, especially when programs are using that connectivity

RealTime Operating system -Example

- A computer system for running an automobile manufacturing assembly line
- The OS receives inputs from sensors along the assembly line…
  - If the OS doesn't satisfy guarantees for processing input data and controlling automated machinery, physical damage will occur
  - If OS misses its timing deadlines: Failure! Halt the assembly line!

Realtime Operating Systems

Real-time operating systems focus on completing tasks by a specific deadline.

Some real-time Operating Systems provide hard real-time guarantees.

- If the OS misses a deadline, this is considered a fatal error!

Embedded Computers

- By far the most common kind of computer now

microwave oven, printer, WiFi router, DVD player, controlling car engine, point-and-shoot camera etc)

- Embedded OSes tend to have very limited capabilities:
  - Systems tend to support a specific, fixed set of tasks
  - Systems aren't designed to run arbitrary programs on them
- Can still include a variety of basic OS capabilities
  - Basic thread-management and scheduling support
  - Basic memory management capabilities
  - Support for software upgrades
  - Support for peripherals like flash cards, USB drives, networking,

Virtual Machines (VMs).

- Modern computers can even run an operating system **as an *application*** within another operating system

- The **host operating system** runs the **guest operating system** as an application

Emulation

A computer with one CPU type simulates another CPU [usually] of a different type,

- allowing applications or even a guest operating system to be run within the host system.

Virtualization

- A computer with one CPU type runs a guest operating system compiled for the same CPU type.
- If the CPU has hardware virtualization support, this will be fast!
- Otherwise, certain CPU features must be emulated by the host OS when running the guest operating system

Hypervisor

- The software that provides a virtual machine for the guest OS is called hypervisor.
  - Enforce isolation between guest operating systems
  - Management of hardware resources shared between guest OSes